Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	*An efficient molding technology for manufacturing high precision infrared wafer-level optics for smart device applications*			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility (√), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing (√), Quantum ( )			
Industrial Technology Classification [1)]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Electrical・Electronics		*Mechanical・Materials*	
	Medium Category	Small Category	Medium Category	Small Category
	Optical Application Devices	*Optical Components*	*Precision Manufacturing Machinery*	*Other Precision Manufacturing Machinery–related Technologies*
Concept of Proposed Technology	The proposed technology focusses on wafer-level glass molding, specifically designed to produce high-precision infrared optics. Wafer-level glass molding is a transformative approach offering significant advantages in cost, efficiency, and glass shape complexity. This innovative method is critical in meeting the rising demand for complex-shaped infrared optics in smart devices across various sectors, such as automotive, healthcare, and consumer electronics.			
Rationale for Joint R&D Project	While wafer-level glass molding enables an efficient and sustainable optics production, developing and ramping-up the entire process chain of such wafer-level technology – including mold manufacturing, fabrication of infrared transmissive chalcogenide glass as well as finally the wafer-level glass molding – is very challenging and requires various very specific capabilities in the individual disciplines.			

(max 10 pages)

	Thus, a close collaboration between experts is needed for successful R&D Project. While Fraunhofer IPT as world-leading institute for glass molding process design and mold manufacturing can focus on the development and demonstration of a wafer-level optics mold, the Korean partner can focus on the development of chalcogenide glass wafers and molding compact infrared lenses.
Key Words	1. *Wafer-level glass molding* 2. *Infrared optics production* 3. *Chalcogenide glass* 4. *Mold manufacturing* 5. *FEM simulations / digital twin*

(max 10 pages)

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (4)	End of the Project	Level (7)

R&D Objectives

The demand for precision infrared optics with complex shapes has risen significantly in recent years (1). This is especially relevant for the automotive industry, including autonomous driving, health care and consumer electronics with immense pressure to maintain competitive pricing for market entry. However, the currently available infrared optics often fall short of the required price range. One effective approach to reduce costs is the replicative production of complex optics through precision glass molding. A key advantage of precision molding is its ability to enhance efficiency by increasing the number of molded lenses produced per cycle, achievable through multiple tools or a wafer-level manufacturing approach.

This implementation approach has the potential to significantly reduce production costs through production scalability, enhancing production times, and minimizing handling over the whole process chain. This applies not only in comparison to traditional direct manufacturing of optical components but also in relation to conventional single-cavity/multi-cavity molding.

Wafer-level glass molding is a highly efficient, but also a very complex manufacturing process. The visco-elastic glass behavior during forming leads to a demanding process design, especially for the wafer-level scale molding. During molding, a non-trivial glass flow takes place in the glass wafer. In addition, complex non-rotationally symmetrical shrinkage behavior occurs in the glass during cooling. On the one hand, this results in shape deviations, and on the other hand, stress peaks can occur in the glass, leading to glass breakage.

This glass behavior is not only considered during glass molding but must already be considered during mold design and mold production. For this, Fraunhofer IPT has developed FEM simulation tools to analyze the viscoelastic glass behavior and predict the glass flow and the shrinkage behavior. By this FEM simulations, Fraunhofer IPT is capable to design and develop optimized mold designs as digital twin, particularly relevant for complex wafer-level molds. Afterwards, Fraunhofer IPT has long-term expertise in developing manufacturing processes for mold fabrication. For mold manufacturing usually ultra-precision (UP) machining as UP grinding, UP turning or UP polishing is applied.

The mold manufactured at Fraunhofer IPT can subsequently be used by the Korean SME to develop an efficient molding process for manufacturing high-precision infrared optics. This R&D works include on one hand the development and fabrication of chalcogenide glass wafers. On the other hand, the final glass molding process must be developed defining crucial process parameters such as the molding temperature, molding force and the cooling parameters.

(max 10 pages)

R&D Target Specifications	The final target specifications have to be discussed with the Korean SME in Phase 1 of the joint R&D Project. The technical requirements for the demonstrator will be defined based on current demand in consumer electronics and automotive sectors and focus on future next-generation infrared sensor requirements. The target resolution of the wafer-level optics is expected to be around 160 × 120. To develop and demonstrate a highly efficient wafer-level, a wafer diameter of around 60 mm could be targeted. The targeted surface qualities for the demonstrator will be based on typical requirements for IR sensors: Peak-to-Valley of mold inserts: < 2 µm, Peak-to-Valley of molded optics: < 4 µm, surface roughness: < 10 nm. At least 25 molded wafer arrays are planned for demonstration.
R&D Plan	The research approach and key tasks involve definition of target specifications, mold design through FEM simulations, mold manufacturing using ultra-precision machining including compensation strategies to improve the achieved mold surface quality, the development of chalcogenide glass wafers including synthesis of chalcogenide glass bulk material and subsequently glass machining, and the development and optimization of the molding process. The key tasks are accompanied by metrological analysis methods such as tactile or optical measurement techniques for analyzing the accuracy of manufactured molds or molded glass wafers. Expected milestones are structured over a 3 year timeline, starting with a requirement analysis in the first month to define target specifications and design criteria (Milestone 1, Month 3). This is followed by mold design completion (Milestone 2, Month 9) and mold fabrication (Milestone 3, Month 18). The development of chalcogenide glass wafers will be performed at the same time (Milestone 4, Month 18). First wafer-level optics could be fabricated by the end of the second year (Milestone 5, Month 24), with subsequent optimization of the mold manufacturing, chalcogenide glass wafer and molding process (Milestone 6, Month 30). Finally, the demonstration phase will span six months, culminating in the production and evaluation of wafer-level optics (Milestone 6, Month 36).

(max 10 pages)

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	**Development and demonstration of a wafer-level optics mold for chalcogenide glass** As the world's leading research institute in mold manufacturing and glass molding technology, Fraunhofer IPT is responsible for the development of a high-precision wafer-level optics mold for the application selected by the Korean SME (2). For this, Fraunhofer IPT will use its longstanding experience and deep understanding of the molding process and the temperature-dependent glass behavior for wafer-level mold design. The process design is carried out using numerical FEM simulations. This numerical modeling - developed at Fraunhofer IPT - allows the prediction of glass flow, shrinkage, index drop, temperature and stress distribution. Afterwards, the physical process development is carried out at Fraunhofer IPT. This includes the mold manufacturing by UP grinding, turning or polishing technology and the evaluation and application of a durable mold coating. Main deliverables are fabricated and coated wafer-level molds which can subsequently be used by the Korean partner for molding process development and wafer-level optics manufacturing.
Expected R&R of Korean SME Partner	**Development of chalcogenide glass wafers and compact infrared (IR) lenses targeting smart device applications** The Korean SME will take the lead in the consortium. As the end user the industry partner is responsible for selecting the optical demonstrator including the optics geometry, required specifications, and the chalcogenide glass type. Furthermore, the Korean partner will develop and demonstrate the wafer-level glass molding process in industrial environments. The molding process must be ramped up by setting up the glass molding facility and develop the glass molding process by adapting process parameters such as the molding temperature, heating and cooling rates, the molding pressure and the molding time. Additionally, the Korean partner is responsible for the development and fabrication of chalcogenide glass wafers. The chalcogenide glass wafers have to achieve the required geometry and surface specifications (e.g. diameter, thickness, surface roughness) for the subsequent molding process. Summarizing, the Korean partner will generate its in-house capabilities for wafer-level molding. The final project outputs are wafer-level glass optics which are fabricated in the Korean partner plant.

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5. Expected Outcomes/Impact

Expected Outcomes and Impact	In this project, a scalable manufacturing process for high-precision infrared lenses will be developed to provide a competitive edge in next-generation glass molding technology. The project will cover the entire process chain of glass molding, and the following outcomes are aimed: • A highly efficient process for wafer-level optics molding • Study of viscoelastic and structural relaxation behavior of glass through molding • Efficient mold manufacturing technology including durable mold protective coatings • Manufacturing of precision chalcogenide glass wafers as preforms for ware-level molding • Competitive supply chain for high-precision, cost-efficient optics / sensors in Korea These outcomes will be demonstrated by the fabrication of wafer-level demonstrators given by the Korean SME. At the end of the project, the Korean partner will be able to carry out own efficient wafer-level molding of high-precision infrared optics. This project therefore directly strengthens the technological and economic situation of the Korean SME. In addition, many further Korean companies that require infrared optical systems will also benefit. In Korea, optical systems and the production of these systems are key components in a wide range of applications such as electronics (Samsung, LG, Hanwha), automotive (Hundai, Kia) or medical technology. These companies utilize optical systems for various applications, including image processing, sensors, displays, and more. Due to the challenges and constantly growing market, the requirement of complex optical components and miniaturized applications demand more efficient and scalable manufacturing solution. Summarizing, for Korean companies, this creates a stronger intra-Korean supply chain, which reduces international dependencies and creates jobs in Korea. Additionally, this strengthens the Korean optical industry in international competition and secure market share in Asia as well as opening new markets, such as Europe.

6. References

(1) https://www.precedenceresearch.com/ir-camera-market

(2) https://www.photonics.com/Articles/Chalcogenide-Glass-Molding-Advances-Precision-IR/a62080

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☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization